

NETWORKS AND COMPLEXITY

Solution 22-3

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 22.3: General two-dimensional system [Lvl. 2]

Consider a network, where the dynamics of the nodes is given by

$$\begin{aligned}\dot{x}_i &= f(x_i, y_i) - c_x \sum L_{ij} x_j \\ \dot{y}_i &= g(x_i, y_i) - c_y \sum L_{ij} y_j\end{aligned}$$

where x_i, y_i are the local variables in node i , f and g are arbitrary functions, $\mathbf{L}$ is the network Laplacian and c_x, c_y are diffusion rates.

- a) Show that, if (x^*, y^*) is the steady state of a network containing only a single node, then every network has a homogeneous steady state at $x_i = x^*, y_i = y^*$ for all i .

Solution

On a single node the system becomes

$$\begin{aligned}\dot{x} &= f(x, y) \\ \dot{y} &= g(x, y)\end{aligned}$$

Hence, a steady state x^*, y^* for the single node system must satisfy $g(x^*, y^*) = 0$ and $f(x^*, y^*) = 0$.

We now consider a state of a system on a network where $x_i = x^*$ and $y_i = y^*$ for all i . Substituting into the first equation of motion yields

$$\dot{x}_i = f(x^*, y^*) - c_x \sum_j L_{i,j} x^* \tag{1}$$

$$= 0 - c_x x^* \sum_j L_{i,j} \tag{2}$$

$$= -c_x x^* \underbrace{\left(k_i - \sum_j A_{i,j} \right)}_0 = 0 \tag{3}$$

And similarly

$$\dot{y}_i = g(x^*, y^*) - c_y \sum_j L_{i,j} y^* \tag{4}$$

$$= 0 - c_y y^* \sum_j L_{i,j} \tag{5}$$

$$= -c_y y^* \left(k_i - \sum_j A_{i,j} \right) = 0 \tag{6}$$

Since none of the variables changes in time, this is a steady state of the entire system.

- b) Compute the Jacobian $\mathbf{J}$ for this system on the linked pair and show that it can be written in the form $\mathbf{J} = \mathbf{I} \otimes \mathbf{P} - \mathbf{L} \otimes \mathbf{C}$, where $\mathbf{P}$ is the Jacobian of a network consisting of a single isolated node.

Solution

On the linked pair we can write the system as

$$\begin{aligned}\dot{x}_1 &= f(x_1, y_1) + c_x(x_2 - x_1) \\ \dot{y}_1 &= g(x_1, y_1) + c_y(y_2 - y_1) \\ \dot{x}_2 &= f(x_2, y_2) + c_x(x_1 - x_2) \\ \dot{y}_2 &= g(x_2, y_2) + c_y(y_1 - y_2)\end{aligned}$$

where we have already substituted the linked-pair Laplacian

$$\mathbf{L} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (7)$$

We can now compute the Jacobian in the usual way

$$\mathbf{J} = \begin{pmatrix} f_x - c_x & f_y & c_x & 0 \\ g_x & g_y - c_y & 0 & c_y \\ c_x & 0 & f_x - c_x & f_y \\ 0 & c_y & g_x & g_y - c_y \end{pmatrix} \quad (8)$$

Now let's compute $\mathbf{J}$ using the shortcut instead. We need the linked pair Laplacian from above, the Jacobian of the single node system

$$\mathbf{P} = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}, \quad (9)$$

the coupling matrix

$$\mathbf{C} = \begin{pmatrix} c_x & 0 \\ 0 & c_y \end{pmatrix}, \quad (10)$$

and the identity matrix of the right size

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (11)$$

Then we can write

$$\mathbf{J} = \mathbf{I} \otimes \mathbf{P} - \mathbf{L} \otimes \mathbf{C} \quad (12)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \mathbf{P} - \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \otimes \mathbf{C} \quad (13)$$

$$= \begin{pmatrix} \mathbf{P} & 0 \\ 0 & \mathbf{P} \end{pmatrix} + \begin{pmatrix} -\mathbf{C} & \mathbf{C} \\ \mathbf{C} & -\mathbf{C} \end{pmatrix} \quad (14)$$

$$= \begin{pmatrix} f_x - c_x & f_y & c_x & 0 \\ g_x & g_y - c_y & 0 & c_y \\ c_x & 0 & f_x - c_x & f_y \\ 0 & c_y & g_x & g_y - c_y \end{pmatrix} \quad (15)$$

which is the same result.